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POINT OF SALE DEVICE WITH CAPACITOR FOR BACKUP POWER

Abstract

A point-of-sale (POS) device may include an input device to receive transaction data, a processor to process the transaction data, a storage memory to store the transaction data, a supercapacitor to provide power to the processor in response to a loss in power, and a voltage monitor to monitor an input voltage to the POS device to identify the loss in power and provide an alert to the processor indicating the loss in power, wherein, in response to the alert, the processor stores the transaction data in the storage memory.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Application No. 63/563,241, filed Mar. 8, 2024, which application is incorporated herein by reference.

BACKGROUND

[0002] Point of sale devices handle transaction data which may be lost due to a power failure. Some point of sale devices include batteries for backup power, but as batteries age, they may lose the ability to hold a charge and/or swell.

SUMMARY

[0003] Various aspects of the disclosure may now be described with regard to certain examples and embodiments, which are intended to illustrate but not limit the disclosure. Although the examples and embodiments described herein may focus on, for the purpose of illustration, specific systems and processes, one of skill in the art may appreciate the examples are illustrative only, and are not intended to be limiting.

[0004] Aspects of the present disclosure are directed to a point-of-sale (POS) device, including an input device to receive transaction data, a processor to process the transaction data, a storage memory to store the transaction data, a supercapacitor to provide power to the processor in response to a loss in power, and a voltage monitor to monitor an input voltage to the POS device to identify the loss in power and provide an alert to the processor indicating the loss in power, wherein, in response to the alert, the processor stores the transaction data in the storage memory.

[0005] In some implementations, the POS device includes a volatile memory, wherein the processor stores the transaction data from the volatile memory to the storage memory. In some implementations, the transaction data includes transaction data which was being written to the storage memory when the loss in power was identified. In some implementations, the transaction data includes cached transaction data in the volatile memory. In some implementations, the processor, in response to the alert indicating the loss in power, shuts down subsystems of the POS device to conserve the power provided by the supercapacitor. In some implementations, the POS device includes a display, wherein shutting down the subsystems of the POS device includes shutting down the display. In some implementations, shutting down the subsystems of the POS device includes shutting down the subsystems of the POS device directly from a kernel of an operating system running on the processor. In some implementations, the processor, in response to the transaction data being stored in the storage memory during the loss in power, shuts down the POS device. In some implementations, the voltage monitor identifies the loss in power by determining that the input voltage has dropped below a predetermined threshold. In some implementations, the processor stores the transaction data in the storage memory within 500 ms of the voltage monitor identifying the loss in power.

[0006] Aspects of the present disclosure are directed to a method, including identifying a loss in input power to a point-of-sale (POS) device, in response to the loss in input power to the POS device, providing power to a processor of the POS device using a supercapacitor, in response to the loss in input power to the POS device, writing, using the processor of the POS device, transaction data from a volatile memory of the POS device to a non-volatile memory of the POS device, and in

response to the transaction data being written to the non-volatile memory, automatically shutting down, using the processor, the POS device.

[0007] In some implementations, identifying the loss in input power includes monitoring an input voltage to the POS device. In some implementations, identifying the loss in input power includes determining that the input voltage has dropped below a predetermined threshold. In some implementations, the transaction data includes transaction data which was being written to the non-volatile memory when the loss in power was identified. In some implementations, the transaction data includes cached transaction data in the volatile memory. In some implementations, in response to the loss in input power to the POS device, shutting down, by the processor, subsystems of the POS device to conserve the power provided by the supercapacitor. In some implementations, shutting down the subsystems of the POS device includes shutting down a display of the POS device. In some implementations, shutting down, by the processor, the subsystems of the POS device includes shutting down the subsystems of the POS device directly from a kernel of an operating system running on the processor. In some implementations, the method includes making, by the processor, a power off API call to the one or more subsystems of the POS device. In some implementations, the processor stores the transaction data to the non-volatile memory within 500 ms of the identification of the loss in power.

[0008] The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features may become apparent by reference to the following drawings and the detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is an example block diagram of a secure point-of-sale (POS) platform.

[0010] FIG. 2 is an example block diagram of components of a POS device including a rechargeable battery.

[0011] FIG. 3 is an example block diagram of components of a POS device including a supercapacitor.

[0012] FIG. 4 is an example block diagram illustrating a system for implementing a graceful shutdown for a POS device.

[0013] FIG. 5 is an example flow chart of a method for implementing a graceful shutdown for a POS device.

[0014] The foregoing and other features of the present disclosure may become apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. Understanding that these drawings depict only several embodiments in accordance with the disclosure and are therefore, not to be considered limiting of its scope, the disclosure may be described with additional specificity and detail through use of the accompanying drawings.

DETAILED DESCRIPTION

[0015] In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here. It may be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the figures, may be arranged, substituted, combined, and designed in a wide variety of different configurations, all of which are explicitly contemplated and made part of this disclosure.

[0016] After a payment transaction is performed at a point-of-sale (POS) device the POS device needs to write transaction data of the payment transaction to a storage memory (i.e., non-volatile memory) of the POS device. In the event of a loss of power to the POS device, the transaction data may be corrupted and/or lost. Power loss may result in loss or corruption of the transaction data, as the POS device could be writing data to the storage memory at the time of the power loss, leading to incomplete write actions and potentially causing data loss or data corruption. This problem is especially severe when POS devices store offline transactions in the POS device without syncing with a payment gateway. When data loss occurs, the offline transactions may be lost, preventing the POS device from submitting the offline transactions to a payment network and leading to a loss of funds for a merchant.

[0017] Some conventional POS devices include a rechargeable battery (e.g., lithium-ion battery) to provide backup power to enable them to complete operations they were performing at the time of the unexpected power loss. However, integrating a lithium-ion battery in a POS device is costly and presents reliability issues over time. The lithium-ion battery requires a battery charger circuit to charge the battery, a gas gauge circuit to track a capacity of the battery, and various safety circuits to keep the device operating safely during charging and discharging.

[0018] The present disclosure provides the advantage of a graceful shutdown of a POS device without use of a battery, and accompanying charger and gas gauge circuits, by providing backup power to the POS device using a supercapacitor of the POS device. The supercapacitor can automatically and immediately provide power to the POS device in response to a loss in power to the POS device. A processor of the POS device can automatically shut down subsystems of the POS device in response to the loss in power to conserve the power provided by the supercapacitor. The processor can shut down the subsystems of the POS device from kernel space to rapidly shut down the subsystems. In this way, the POS device can execute a rapid and graceful shutdown without loss or corruption of transaction data in response to a loss in power.

[0019] As shown in FIG. 1, secure POS platform **100** may include a number of hardware components including input device controller (IDC) **102**, multiplexer (MUX) **104**, secure processor (SP) **106**, and applications processor (AP) **108**. Throughout this specification, a touch screen will be used as an exemplary input device such that the input device controller may be referred to as a touch controller (TC). However, as mentioned previously, the input device controller could be a controller for any number of input devices or user interfaces such as a keypad, audio interface, or wireless interface. The above are only examples of components that may be included in a secure POS platform, such as **100**, and other components may also be included.

[0020] Referring to FIG. 1, there is shown an example schematic view of a secure POS platform **100** in accordance with one or more example embodiments. In FIG. 1, secure POS platform **100** can include one or more components, such as **102**, **104**, **106**, and **108** that are internal to associated client device **120**, which can be a merchant POS device. Client device **120** may be operated by a user **110** entering various information for a purchase transaction via an associated input device associated with the client device **120** to facilitate a purchase transaction. The input device could be a touch screen integrated with the client device **120**. The input device could be a multiuse input device such that it is used for both secure operations and normal (non-secure) operations. Secure POS platform **100** may evaluate whether the certain user inputs via the multiuse input device are associated with either secure operations or normal (non-secure) operations. An example of a secure operation is entering a PIN. An example of a non-secure operation is entering a quantity of an item to be purchased via the same touch screen.

[0021] The hardware components, such as **102**, **104**, and **106**, may be interconnected to provide various functions in response to certain actions performed by user **110** with respect to client device **120**, such as inputting information via an associated input device or user interface. The hardware components can be implemented as individual integrated circuits (ICs) on a main board of client device **120**. However, MUX **104** and SP **106** could also be implemented on a single application

specific integrated circuit (ASIC) while IDC **102** and AP **108** are kept as separate ICs. The chip comprising MUX **104** and SP **106** could then be used as a “secure chip” in a chip set comprising the secure chip, IDC **102**, and AP **108**. As will be described later, integrating MUX **104** and SP **106** onto a single chip provides certain benefits in that IDC **102** and AP **108** could be commercially available ICs that are broadly available beyond the more limited market for specialized secure POS hardware. In other words, an application processor chip and peripheral controller chip could be standard devices for the control of peripherals and running applications generally, and do not need to be augmented in any manner to realize the security benefits of some of the approaches described herein. Any of the functions performed by SP **106** that are disclosed in this specification could be executed on the secure chip. For example, tamper detection circuitry implemented on the secure chip could receive a tamper indication from a tamper sensor and clear a memory implemented on the secure chip to protect encryption keys or other secure information stored on the secure chip.

[0022] AP **108** can be a standard processor used to implement an operating system for client device **120**. For example, AP **108** can run the Android OS™. In general, the AP can maintain full control over the client device **120** while the device is running applications that don't require secure information. For example, the main application run by the AP **108** could be the standard merchant-facing retail checkout application used to accept inputs regarding a customer's order and tally the total of a specific purchase transaction. As another example, the main application run by AP **108** could be a restaurant server's ordering application used to input orders taken by the server. AP **108** can operate with a cache and main memory that are physically separate from the memory used by SP **106**. This configuration provides certain benefits in that certain attacks may use a processor to write instructions to memory for later execution and exploitation of a secure program running on the same processor, or may use the processor to read and illicitly obtain data from the memory which was utilized and left over by a secure program using the same memory. These kinds of vulnerabilities can be avoided by not allowing the AP to have access to the same memory as the SP. MUX **104** is a hardware multiplexer that receives control inputs from SP **106** and routes data received from IDC **102** to one or more data channels based on those control inputs. The data could be routed through transistors that turn on or off in response to control signals. MUX **104** could be an IC with one or more control pins, one or more input pins and at least two output pins. In approaches where MUX **104** and SP **106** were implemented on a single chip, MUX **104** could be a block of analog circuitry on the secure chip controlled by a digital output from SP **106**. In such approaches, MUX **104** and SP **106** could exchange information via interconnects in the IC.

[0023] In some embodiments, illustrated by the region of FIG. **1** indicated by reference number **10**, a secure transaction process can be implemented using the hardware components shown in the portion of FIG. **1** indicated by reference number **10**. For example, when user **110** interacts with an input device or user interface associated with client device **120**, IDC **102** can transmit certain user inputs to MUX **104**. In certain instances, when secure POS platform **100** detects or prompts user entry of secure data (e.g., a user entering PIN information) via the input device or user interface associated with client device **120**, the operation is treated as a secure transaction and/or operation. In this instance, SP **106** can control the flow of secure data through MUX **104**.

[0024] FIG. **1** illustrates another example schematic view of a secure POS platform which is illustrated by the region of FIG. **1** indicated by reference number **20**, which is in accordance with one or more example embodiments. In the embodiments indicated by reference number **20**, entry of data that may be considered non-secure (normal mode) can be facilitated via the hardware components, such as **102**, **104**, **106**, and **108** of client device **120**, such that data passes from a peripheral to AP **108**. For example, IDC **102** may communicate user inputs to MUX **104**, such as when user **110** interacts with an associated input device or user interface. MUX **104** can then pass the information received from this interaction to AP **108**. AP **108** may then perform transactions and/or operations that may be considered non-secure. In addition, entry of data that may be considered secure can be facilitated via the same hardware components, such as **102**, **104**, **106**, and

108 of client device **120**. However, the data from the peripheral will now be routed to SP **106** instead of AP **108**. For example, under the control of SP **106**, MUX **104** will route input data from the peripheral to SP **106**. SP **106** may then perform operations on the data that may be considered secure. IDC **102** may be completely unaware as to which processor the input data is being routed to.

[0025] The secure POS platform **100** may be a secure POS platform as described in U.S. Pat. No. 11,393,300, which patent is incorporated herein by reference in its entirety.

[0026] FIG. **2** is an example block diagram of components of a POS device **200** including a rechargeable battery **280**. The POS device **200** may include a battery charger **260** which receives an input power **290** from an external source, such as a wall outlet. The battery charger **260** may charge the battery **280** using the input power **290**. A gas gauge circuit **270** may monitor a charge of the battery **280** and control the battery charger **260** to charge the battery **280**.

[0027] The POS device **200** may include voltage regulators **250** which receive the input power **290** and/or power from the battery **280**. The battery **280** may provide power to the voltage regulators **250** in response to a loss of the input power **290**. In an example, in the event of a power outage, causing the input power **290** to cease, the battery **280** may provide power to the voltage regulators **250**. The voltage regulators **250** may regulate the input power **290** and/or the power from the battery **280** to provide a voltage supply **255**.

[0028] The POS device may include a processor **210**, a volatile memory **220**, a non-volatile memory **230**, and one or more subsystems **240**. The voltage regulators **250** may provide the voltage supply **255** to the processor **210**, the volatile memory **220**, the non-volatile memory **230**, and the one or more subsystems **240**. The voltage supply **255** may be different for one or more of the processor **210**, the volatile memory **220**, the non-volatile memory **230**, and the one or more subsystems **240**. In an example, each of the processor **210**, the volatile memory **220**, the non-volatile memory **230**, and the one or more subsystems **240** receive a different voltage supply **255** from the voltage regulators **250**.

[0029] In some implementations, the processor **210** is an applications processor of the POS device **200**, such as the applications processor **108** of FIG. **1**. The processor **210** may be coupled to the volatile memory **220**, the non-volatile memory **230**, and the one or more subsystems **240**. The processor **210** may control data writes to the volatile memory **220** and the non-volatile memory **230**. The processor **210** may receive information from and/or control the one or more subsystems **240**. The one or more subsystems **240** may include a display, an input device, a printer, a user interface, and other subsystems of the POS device **200**. In an example, the POS device **200** includes a display and the processor **210** causes the display to present a user interface to a user. In an example, the POS device **200** includes a keypad and a display and the processor **210** causes the display to prompt user input at the keypad. In an example, the POS device **200** includes a touchscreen representing a display and input device of the POS device **200** controlled by the processor **210**.

[0030] FIG. **3** is an example block diagram of components of a POS device **300** including a supercapacitor **380**. The POS device **300** may receive an input power **390** from an external source, such as a wall outlet. The supercapacitor **380** may be charged using the input power **390**. The supercapacitor **380** may also be referred to as an ultracapacitor. In some implementations, the supercapacitor **380** may store energy using electrostatic double-layer capacitance and/or electrochemical pseudocapacitance. In an example, the supercapacitor **380** includes two electrodes separated by an ion-permeable membrane and an electrolyte ionically connecting the two electrodes.

[0031] The POS device **300** may include voltage regulators **350** which receive the input power **390** and/or power from the supercapacitor **380**. The supercapacitor **380** may provide power to the voltage regulators **350** in response to a loss of the input power **390**. In an example, in the event of a power outage, causing the input power **390** to cease, the supercapacitor **380** may provide power to

the voltage regulators **350**. The voltage regulators **350** may regulate the input power **390** and/or the power from the supercapacitor **380** to provide a voltage supply **355**.

[0032] The POS device may include a processor **310**, a volatile memory **320**, a non-volatile memory **330**, and one or more subsystems **340**. The voltage regulators **350** may provide the voltage supply **355** to the processor **310**, the volatile memory **320**, the non-volatile memory **330**, and the one or more subsystems **340**. The voltage supply **355** may be different for one or more of the processor **310**, the volatile memory **320**, the non-volatile memory **330**, and the one or more subsystems **340**. In an example, each of the processor **310**, the volatile memory **320**, the non-volatile memory **330**, and the one or more subsystems **340** receive a different voltage supply **355** from the voltage regulators **350**.

[0033] In some implementations, the processor **310** is an applications processor of the POS device **300**, such as the applications processor **108** of FIG. **1**. The processor **310** may be coupled to the volatile memory **320**, the non-volatile memory **330**, and the one or more subsystems **340**. The processor **310** may control data writes to the volatile memory **320** and the non-volatile memory **330**. The processor **310** may receive information from and/or control the one or more subsystems **340**. The one or more subsystems **340** may include a display, an input device, a printer, a user interface, and other subsystems of the POS device **300**. In an example, the POS device **300** includes a display and the processor **310** causes the display to present a user interface to a user. In an example, the POS device **300** includes a keypad and a display and the processor **310** causes the display to prompt user input at the keypad. In an example, the POS device **300** includes a touchscreen representing a display and input device of the POS device **300** controlled by the processor **310**.

[0034] The volatile memory **320** may be a memory requiring power to store data. In an example, the volatile memory **320** is double data rate (DDR) synchronous dynamic random-access memory (SDRAM). In an example, the volatile memory **320** is random-access memory (RAM). The volatile memory **320** may be coupled to a memory controller for the volatile memory **320** configured to write data to and from the volatile memory **320**. In some implementations, the memory controller for the volatile memory **320** is integrated into the processor **310**. The non-volatile memory **330** may be a memory which retains data after power is lost, otherwise referred to as “storage memory.” In an example, the non-volatile memory **330** is an embedded multimedia card (eMMC). In an example, the non-volatile memory **330** is a hard drive, a secure digital (SD) drive, or a solid-state drive (SDD). The non-volatile memory **330** may be coupled to a memory controller for the non-volatile memory **330** configured to write data to and from the non-volatile memory **330**. In some implementations, the memory controller for the non-volatile memory **330** is integrated into the processor **310**.

[0035] The processor **310** may process the transaction data to approve or deny the transaction. In some implementations, the processor **310** may approve or deny transactions in an offline mode, where transaction approval cannot be obtained over payment networks. In the offline mode, the processor **310** may cache offline transaction data in the volatile memory **320** and/or store the offline transaction data in the non-volatile memory **330** for transaction approval once the POS device **300** is able to access the payment networks. In an example, the POS device **300** may be used to process transactions in a national park without a network connection and the POS device **300** stores the offline transaction data in the non-volatile memory **330** until the POS device **300** is taken to a city near the national park where a network connection can be established to provide the offline transaction data to the payment networks for approval. When transactions are being processed, and/or when transaction data is cached in the volatile memory **320**, there is a risk that the transaction data may be lost if the input power **390** is suddenly lost. The transaction data of a transaction being processed when the input power **390** is lost may be corrupted if the processor **310** is in the process of writing the transaction data to the non-volatile memory **330**. Transaction data that is cached in the volatile memory **320** may be lost when the input power **390** is lost.

[0036] To prevent loss or corruption of transaction data, the POS device **300** includes the supercapacitor **380** which provides power after a loss of the input power **390** to allow the transaction data to be written to the non-volatile memory **330**. The supercapacitor **380** may be charged using the input power **390** in order to provide power after the input power **390** is lost. The supercapacitor may store a predetermined amount of power to allow the POS device **300** to perform a graceful shutdown (i.e., a shutdown without lost or corrupted data).

[0037] The POS device **300** may include an input voltage monitor **370** to monitor a voltage of the input power **390**. The input voltage monitor **370** may identify a loss of the input power **390** based on the voltage of the input power **390** dropping below a predetermined threshold. The input voltage monitor **370** may send an alert to the processor **310** in response to the loss of the input power **390**. The alert to the processor **310** may indicate the loss of the input power **390**. In some implementations, the alert to the processor **310** indicates that the supercapacitor **380** is providing power. In some implementations, the alert to the processor **310** includes an amount of time the supercapacitor can provide power or an amount of power the supercapacitor can provide. In some implementations, the alert may include a hardware interrupt **375**. The hardware interrupt **375** may interrupt one or more processes of the processor **310**.

[0038] In response to the loss of the input power **390**, the supercapacitor **380** may automatically provide power to the voltage regulators **350**. In some implementations, the POS device includes a switch circuit **360** to switch from providing power from the input power **390** to providing power from the supercapacitor **380**. In some implementations, the switch circuit **360** includes circuitry to automatically draw power from the supercapacitor **380** in response to the loss of the input power **390**.

[0039] The processor **310**, in response to the alert from the input voltage monitor **370** indicating the loss of the input power **390**, may shut down the one or more subsystems **340** of the POS device **300**. The processor **310** may shut down the one or more subsystems **340** of the POS device **300** to conserve the power supplied by the supercapacitor **380**. The one or more subsystems **340** may include a display subsystem including a display. Shutting down the one or more subsystems **340** may include shutting down the display and/or a graphics processor for the display. Shutting down the one or more subsystems **340** may be performed by a kernel of an operating system running on the processor **310**. By shutting down the one or more subsystems **340** using the kernel, or in kernel space, the processor **310** may shut down the one or more subsystems more quickly than shutting down the one or more subsystems **340** in user space. In this way, the processor **310** may conserve the power supplied by the supercapacitor **380**, allowing for the supercapacitor **380** to be smaller while still allowing the processor **310** to execute a graceful shutdown.

[0040] In some implementations, shutting down the one or more subsystems may include sending, by the processor **310**, a shutdown command to the one or more subsystems. In an example, the processor **310** makes one or more power off API calls to the one or more subsystems. In some implementations, shutting down the one or more subsystems may include turning off the power supply to the one or more subsystems. In an example, the processor **310** turns off the voltage regulators that supply power to the one or more subsystems. In some implementations, shutting down the one or more subsystems may include physically disconnecting the one or more subsystems from the power provided by the supercapacitor **380**.

[0041] The supercapacitor **380** may supply power to the processor **310**, the volatile memory **320**, and the non-volatile memory **330** to allow the processor **310** to write transaction data from the volatile memory **320** to the non-volatile memory **330**. In some implementations, the processor **310** completes an in-progress write to the non-volatile memory **330**. In some implementations, the processor **310** transfers cached transaction data in the volatile memory **320** to the non-volatile memory **330**. The supercapacitor **380** may store an amount of power to allow the processor **310** to complete all write operations from the volatile memory **320** to the non-volatile memory **330**.

[0042] In response to the transaction data being stored in the non-volatile memory **330**, or in

response to the processor **310** completing all the write operations from the volatile memory **320** to the non-volatile memory **330**, the processor **310** may shut down the POS device **300**. In this way, the POS device **300** is not shut down until the transaction data is moved from the volatile memory **320** to the non-volatile memory **330**, using power provided by the supercapacitor **380** and preventing corruption and/or loss of the transaction data. This shutdown may be termed a graceful shutdown, as the transaction data is not lost or corrupted during shutdown. In some implementations, the processor **310** may shut down the POS device **300** and complete the graceful shutdown within 500 ms of the notification of the loss of the input power **390**. The supercapacitor **380** may provide power to the processor **310** for an amount of time sufficient to complete the graceful shutdown. The supercapacitor **380** may provide power to the processor **310**, the volatile memory **320**, and the non-volatile memory **330** for the amount of time sufficient to complete the graceful shutdown, or to write the transaction data to the non-volatile memory **330**. In some implementations, the supercapacitor **380** provides power for about 220 ms to one second after the loss in power to the POS device **300**, which is sufficient time to complete the graceful shutdown, or to write the transaction data to the non-volatile memory **330**.

[0043] FIG. **4** is an example block diagram illustrating a system **400** for implementing a graceful shutdown for a POS device. The system **400** may include one or more hardware and/or software components. The system **400** may include a hardware level **410**, a kernel space **420**, and a user space **430**. The hardware level **410** may include a hardware interrupt **475**. The hardware interrupt may be received by a processor of a POS device from a voltage monitor, such as the hardware interrupt **375** of FIG. **3**. A power management integrated circuit (PMIC) charger driver **421** may listen to the hardware interrupt **475**. The PMIC charger driver **421**, in response to the hardware interrupt **475**, informs a power supply framework **422** of the change in power state (i.e., loss of input power). The power supply framework **422** notifies a reboot driver **424** of the change in power state. In some implementations, the power supply framework **422** informs a power manager **431** in the user space **430** of the change in power state. However, waiting for the power manager **431** to shut down subsystems of the POS device takes longer than shutting down the subsystems from the kernel space. The reboot driver **424** initiates a shutdown process directly from the kernel space by calling power off functions provided by a kernel shutdown syscall **425**. The kernel shutdown syscall **425** may be a kernel system call to shut down. The kernel shutdown syscall **425** may send one or more shutdown commands to power off subsystems of the POS device. The subsystems of the POS device may register for a power off notification. The one or more shutdown commands from the kernel shutdown syscall **425** may include the power off notification to shut down the subsystems which register for the power off notification.

[0044] The kernel shutdown syscall **425** notifies a non-volatile memory driver **426** of the shutdown process. In some implementations, the non-volatile memory driver **426** is an eMMC core driver. The non-volatile memory driver **426** sends a shutdown notification to a non-volatile memory controller **411**. In some implementations, the non-volatile memory controller **411** is an eMMC card controller. The non-volatile memory controller **411** flushes data currently being transferred (in-flight data) and cached data to the non-volatile memory (e.g., eMMC). In an example, the non-volatile memory controller **411** sends a POWER_OFF_SHORT notification to the non-volatile memory for a faster shutdown. In this example, the non-volatile memory may complete the data and shutdown operations within 255 ms.

[0045] The kernel shutdown syscall **425** ends the graceful shutdown by powering off the POS device. The kernel shutdown syscall **425** may power off the POS device based on the non-volatile memory completing the data and shutdown operations. The kernel shutdown syscall **425** may power off the POS device based on an amount of time elapsing since the kernel shutdown syscall **425** notified the non-volatile memory driver **426** of the shutdown process.

[0046] FIG. **5** is an example flow chart of a method **500** for implementing a graceful shutdown for a POS device. The method **500** may include more, fewer, or different operations than shown. The

operations may be performed in the order shown, in a different order, or concurrently. The method **500** may be performed by the POS **300** of FIG. **3**.

[0047] At operation **510**, a loss in power to a POS device is identified. In some implementations, identifying the loss in power includes monitoring an input voltage to the POS device. In some implementations, identifying the loss in input power includes determining that the input voltage has dropped below a predetermined threshold.

[0048] At operation **520**, in response to the loss in input power to the POS device, power is provided to a processor of the POS device using a supercapacitor.

[0049] In some implementations, the method **500** includes, in response to the loss in input power to the POS device, shutting down, by the processor, subsystems of the POS device to conserve the power provided by the supercapacitor. In some implementations, shutting down the subsystems of the POS device includes shutting down a display of the POS device. In some implementations, shutting down, by the processor, the subsystems of the POS device includes shutting down the subsystems of the POS device directly from a kernel of an operating system running on the processor. In some implementations, the method **500** includes making, by the processor of the POS device, a power off API call to the one or more subsystems of the POS device.

[0050] At operation **530**, in response to the loss of input power to the POS device, transaction data is written, using the processor of the POS device, from a volatile memory of the POS device to a non-volatile memory of the POS device. In some implementations, the transaction data includes transaction data which was being written to the storage memory when the loss in power was identified. In some implementations, the transaction data includes cached transaction data in the volatile memory. In some implementations, the processor stores the transaction data in the non-volatile memory within 500 ms of the identification of the loss in power.

[0051] In some implementations, the processor shuts down the one or more subsystems before completing the write operations. In some implementations, the processor completes the write operations before shutting down the one or more subsystems. In some implementations, the processor begins the write operations before beginning to shut down the one or more subsystems. In some implementations, the processor begins to shut down the one or more subsystems before beginning the write operations.

[0052] At operation **540**, in response to the transaction data being written to the non-volatile memory, the POS device is automatically shut down using the processor.

[0053] The various illustrative logical blocks, circuits, modules, routines, and algorithm steps described in connection with the embodiments disclosed herein can be implemented as electronic hardware, or combinations of electronic hardware and computer software. To clearly illustrate this interchangeability, various illustrative components, blocks, modules, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware, or as software that runs on hardware, depends upon the particular application and design constraints imposed on the overall system. The described functionality can be implemented in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the disclosure.

[0054] Moreover, the various illustrative logical blocks and modules described in connection with the embodiments disclosed herein can be implemented or performed by a machine, such as a general purpose processor device, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A control processor can synthesize a model for an FPGA. For example, the control processor can synthesize a model for logical programmable gates to implement a tensor array and/or a pixel array. The control channel can synthesize a model to connect the tensor array and/or pixel array on an FPGA, a reconfigurable chip and/or die, and/or the like. A general purpose processor device can be a microprocessor, but in the alternative, the

processor device can be a controller, microcontroller, or state machine, combinations of the same, or the like. A processor device can include electrical circuitry configured to process computer-executable instructions. In another embodiment, a processor device includes an FPGA or other programmable device that performs logic operations without processing computer-executable instructions. A processor device can also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. Although described herein primarily with respect to digital technology, a processor device may also include primarily analog components. For example, some or all of the algorithms described herein may be implemented in analog circuitry or mixed analog and digital circuitry. A computing environment can include any type of computer system, including, but not limited to, a computer system based on a microprocessor, a mainframe computer, a digital signal processor, a portable computing device, a device controller, or a computational engine within an appliance, to name a few.

[0055] The elements of a method, process, routine, or algorithm described in connection with the embodiments disclosed herein can be embodied directly in hardware, in a software module executed by a processor device, or in a combination of the two. A software module can reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other form of a non-transitory computer-readable storage medium. An exemplary storage medium can be coupled to the processor device such that the processor device can read information from, and write information to, the storage medium. In the alternative, the storage medium can be integral to the processor device. The processor device and the storage medium can reside in an ASIC. The ASIC can reside in a user terminal. In the alternative, the processor device and the storage medium can reside as discrete components in a user terminal.

[0056] Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without other input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list.

[0057] While the above detailed description has shown, described, and pointed out novel features as applied to various embodiments, it can be understood that various omissions, substitutions, and changes in the form and details of the devices or algorithms illustrated can be made without departing from the spirit of the disclosure. As can be recognized, certain embodiments described herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others.

[0058] The herein described subject matter sometimes illustrates different components contained within, or connected with, different other components. It is to be understood that such depicted architectures are merely exemplary, and that in fact many other architectures can be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively “associated” such that the desired functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the desired functionality is achieved, irrespective of

architectures or intermedial components. Likewise, any two components so associated can also be viewed as being “operably connected,” or “operably coupled,” to each other to achieve the desired functionality, and any two components capable of being so associated can also be viewed as being “operably couplable,” to each other to achieve the desired functionality. Specific examples of operably couplable include but are not limited to physically mateable and/or physically interacting components and/or wirelessly interactable and/or wirelessly interacting components and/or logically interacting and/or logically interactable components.

[0059] With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations may be expressly set forth herein for sake of clarity.

[0060] It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to inventions containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances, where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.” Further, unless otherwise noted, the use of the words “approximate,” “about,” “around,” “substantially,” etc., mean plus or minus ten percent.

[0061] The foregoing description of illustrative embodiments has been presented for purposes of illustration and of description. It is not intended to be exhaustive or limiting with respect to the precise form disclosed, and modifications and variations are possible in light of the above teachings or may be acquired from practice of the disclosed embodiments. It is intended that the scope of the invention be defined by the claims appended hereto and their equivalents.

Claims

1. A point-of-sale (POS) device, comprising: an input device configured to receive transaction data; a processor configured to process the transaction data; a memory controller configured to store the transaction data in a non-volatile memory; a voltage monitor configured to monitor an input voltage to the POS device from an external power source to identify a loss in power to the POS device and provide an alert to the processor indicating the loss in power; and a supercapacitor configured to provide power to the processor and memory controller in response to the alert, wherein the supercapacitor provides power for an amount of time sufficient to store the transaction data in the non-volatile memory.
2. The POS device of claim 1, further comprising a volatile memory, wherein the processor causes the transaction data to be moved from the volatile memory to the non-volatile memory.
3. The POS device of claim 2, wherein the transaction data includes transaction data which was being written to the non-volatile memory when the loss in power was identified.
4. The POS device of claim 1, wherein the transaction data includes cached transaction data in the volatile memory.
5. The POS device of claim 1, wherein the processor, in response to the alert indicating the loss in power, shuts down subsystems of the POS device to conserve the power provided by the supercapacitor.
6. The POS device of claim 5, further comprising a display, wherein shutting down the subsystems of the POS device includes shutting down the display.
7. The POS device of claim 5, wherein shutting down the subsystems of the POS device includes shutting down the subsystems of the POS device directly from a kernel of an operating system running on the processor.
8. The POS device of claim 1, wherein the processor, in response to the transaction data being stored in the storage memory during the loss in power, shuts down the POS device.
9. The POS device of claim 1, wherein the voltage monitor identifies the loss in power by determining that the input voltage has dropped below a predetermined threshold.
10. The POS device of claim 1, wherein the transaction data is stored in the non-volatile memory within 500 ms of the voltage monitor identifying the loss in power.
11. A method, comprising: identifying a loss in input power from an external power source to a point-of-sale (POS) device; in response to the loss in input power to the POS device, providing power to a processor and memory controller of the POS device using a supercapacitor; in response to the loss in input power to the POS device, writing, using the processor of the POS device, transaction data from a volatile memory of the POS device to a non-volatile memory of the POS device, wherein the supercapacitor provides power for an amount of time sufficient to write the transaction data to the non-volatile memory; and in response to the transaction data being written to the non-volatile memory, automatically shutting down, using the processor, the POS device.
12. The method of claim 11, wherein identifying the loss in input power includes monitoring a voltage of the input power to the POS device.
13. The method of any of claim 12, wherein identifying the loss in input power includes determining that the voltage has dropped below a predetermined threshold.
14. The method of claim 11, wherein the transaction data includes transaction data which was being written to the non-volatile memory when the loss in power was identified.
15. The method of claim 11, wherein the transaction data includes cached transaction data in the volatile memory.
16. The method of claim 15, further comprising, in response to the loss in input power to the POS device, shutting down, by the processor, subsystems of the POS device to conserve the power provided by the supercapacitor.

17. The method of claim 16, wherein shutting down the subsystems of the POS device includes shutting down a display of the POS device.

18. The method of claim 16, wherein shutting down, by the processor, the subsystems of the POS device includes shutting down the subsystems of the POS device directly from a kernel of an operating system running on the processor.

19. The method of claim 18, further comprising making, by the processor, a power off API call to the subsystems of the POS device.

20. The method of claim 11, wherein the transaction data is written to the non-volatile memory within 500 ms of identifying the loss in power.
